PROJECT DEFINITION DOCUMENT

Cross-Platform Content Optimisation AI Tool

Creator Compass

Degree Program: Computer Science BSc
Consultant: Abigail Parisaca Vargas
Project proposed by: Sade Onitiri-Coker
Arrangements for Proprietary Interests: This is not a client-based project, so this does not apply
Promises Made to Secure Acceptance of the Project: None

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Word Count: 1672

PROPOSAL

PROBLEM TO BE SOLVED

In today's digital society, content creation has become both highly competitive and data driven. The short-form content industry, dominated by platforms such as TikTok, Instagram Reels and YouTube Shorts, has reshaped how visibility and engagement are achieved (Manic, 2024).Current solutions in the market primarily focus on retrospective analytics, often limited to basic engagement metrics such as views, likes and shares which provide little actionable insights into why some content performs better than others (Varga and Cucu, 2025). As a result, content creators and social media managers are forced to rely on intuition, imitation of viral trends and trial and error experimentation when it comes to coming up with video ideas.

Moreover, existing AI tools for content optimisation tend to act as "black boxes" offering recommendations without transparency or interpretability (Hase and Bansal, 2020).This lack of explainability, reduces user trust and limits the ability of creators to make informed decisions.

This project addresses this gap by developing an AI-driven, explainable content ideation and optimisation system that assists creators throughout the pre-publication stage. The system will identify emerging trends and opportunities, optimise user-generated or AI generated content ideas and explain the reasoning behind each recommendation. The system will make use of a combination of natural language processing (NLP) for trend detection and semantic analysis, deep learning models for engagement prediction, and explainable AI (XAI) techniques such as SHAP (Shapley Additive Explanations) to interpret model behaviour (Mosca *et al.*, 2022). This will ensure that users receive transparent, data-backed insight rather than opaque algorithmic suggestions.

Ultimately, this project seeks to bridge the gap between complex AI systems and practical creative decision making, enabling content creators to make well informed, data-driven choices while maintaining transparency and agency, The result will be a tool that transforms intuition-based workflows into a situation, explainable and adaptive optimisation process for the modern creator industry.

PROJECT OBJECTIVES

Main Objective: The main objective of this project is to develop an AI-driven, explainable content ideation and optimisation system that assists content creators in generating, refining and understanding short form video ideas based on real-time social media trends.

The Sub-Objectives required for this project are:

1) Creator Compass shall collect and analyse trending topics from the Twitter(X) API and other open datasets using natural language processing (NLP) modelling techniques.
   Test: Verify that the system successfully retrieves trending data and identifies emerging topics within chosen niche.
2) Creator Compass shall generate tailored short-form content ideas by applying NLP models to extracted trend topics, aligning with user provided niches or preferences.
   Test: Verify that each generated output is coherent and contextually relevant and aligned with user intent.
3) Creator Compass shall generate actionable recommendations to improve engagement potential for the user's content ideas both user generated and AI generated.
   Test: Confirm optimisation suggestions such as improved captions, hashtag selection and keyword refinement are generated for each idea.
4) Creator Compass shall integrate explainable AI (XAI) methods such as SHAP and LIME to interpret model outputs showing which features most strongly influenced each recommendation.
   Test: Explanations are correctly generated and correspond to model outputs.
5) Creator Compass shall implement a user-friendly interface that integrates all modules into a unified dashboard.
   Test: Verify that the frontend successfully communicates with all backend components and presents information clearly.

After these core objectives have been met, some of these additional features will enhance the systems functionality:

6) Creator Compass shall implement a user profiling system to adapt future recommendations based on a creator past content ideas and analysis history.
   Test: Verify that historical user data is correctly stored, retrieved and used to influence future suggestions.
7) Creator Compass shall analyse activity patterns and predict optimal posting times for each platform to create a predictive posting schedular.
   Test: Posting time recommendations are generated using platform trend and engagement data and validated against past post-performance patterns.

PROJECT BENEFICIARIES

Individuals or organisations who may benefit from Creator Compass include:

- Content Creators and Influencers: They will gain actionable insights into what makes posts perform well across platforms. The tool reduces guesswork by offering transparent recommendations on content ideas, hashtags and creative style, helping creators to make informed decisions that improve engagement and consistency.
- Digital Marketing Professionals and Agencies: They can analyse engagement trends more effectively and understand why certain creative elements perform better, improving campaign strategy and client reporting.
- Researchers and Developers in Explainable AI (XAI): They can study how interpretability methods can be applied to real world social media prediction problems, especially where transparency for non-technical users is crucial.

PROJECT WORK PLAN

I plan on using Agile methodology to deliver Creator Compass. All actions and decisions will be documented, and project progress will be managed in Notion to track tasks, sprints and deliverables. The project is structured into four phases, with each phase containing 3 two-week sprints with clear deliverables. Development will focus on three main modules:

- M1: Trend Analysis & Ideation
- M2: Optimisation
- M3: Explainability

Phase 1: Environment Setup & Technology Exploration (Sprint 1-3: 13 Oct -23 Nov 2025)
In this phase, I will finalise the tech stack and development environment. The backend will be implemented using Python with Flask/FastAPI, and the UI will be built using Streamlit, chosen for their efficiency and ease of integration. SHAP and LIME are selected for their compatibly with deep learning models and ability to provide clear, interpretable insights. Initial tests on potential datasets will be conducted, including sample API calls and early preprocessing scripts. The deliverable for this phase will be a functional development environment and initial tests scripts.

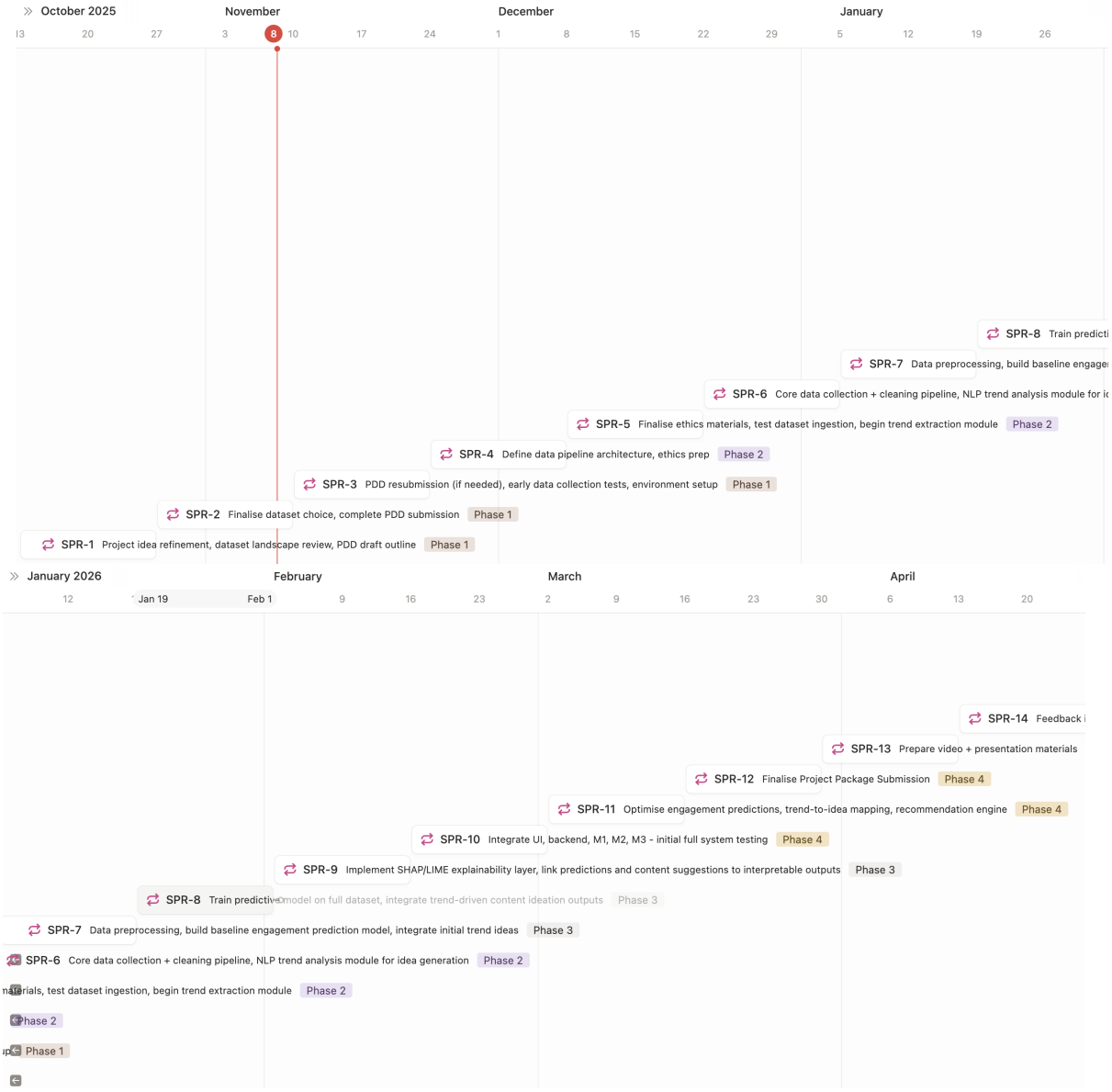
Phase 2: Data Collection & Preprocessing (Sprint 4-6: 24 Nov 2025– 4 Jan 2026)
This phase focuses on collecting social media trend data, building the data pipelines and preparing the datasets, using EDA, for both engagement prediction and trend-based content ideation. I will also develop preprocessing modules for metadata inputs. The deliverable at this stage will be ready to use, cleaned datasets, trend extraction scripts and preprocessing pipelines.

Phase 3: Model Development & Explainability (Sprint 7-9: 5 Jan -15 Feb 2026)
In this phase I will develop the predictive and explainable AI components, integrating all three modules. Models will be trained on engagement data and trend-based idea generation will be linked to the logic. XAI techniques will be applied to interpret predictions. The deliverable will be a working predictive model with explainability features, a trend driven content ideation module and initial recommendation logic.

Phase 4: System Integration & Optimisation (Sprint 10-12: 16 Feb -29 Mar 2026)
This phase focuses on integrating all modules into a fully functional cross-platform tool. The UI, backend, predictive model, trend analysis module, explainability and recommendation engine will be combined. Full system testing will be performed to validate both performance and usability.

The deliverable for this phase will be a fully integrated prototype that has been well tested and detailed documentations of the system.

Sprint	Duration	Phase	Key Deliverable	Target Effort (Hours)	Is Current/Next
SPR-1	October 13, 2025 → October 26, 2025	Phase 1	Project idea refinement, dataset landscape review, PDD draft outline	25	⬅️ PAST
SPR-	October 27, 2025 → November 9, 202	Phase 1	Finalise dataset choice, complete PDD submission	30	✅ CURRENT
SPR-3	November 10, 2025 → November 23, 2025	Phase 1	PDD resubmission (if needed), early data collection tests, environment setup	35	➡️ NEXT
SPR-4	November 24, 2025 → December 7, 2025	Phase 2	Define data pipeline architecture, ethics prep	30	FUTURE
SPR-5	December 8, 2025 → December 21, 2025	Phase 2	Finalise ethics materials, test dataset ingestion, begin trend extraction modul	25	FUTURE
SPR-6	December 22, 2025 → January 4, 2026	Phase 2	Core data collection + cleaning pipeline, NLP trend analysis module for idea g	25	FUTURE
SPR-7	January 5, 2026 → January 18, 2026	Phase 3	Data preprocessing, build baseline engagement prediction model, integrate i	35	FUTURE
SPR-8	January 19, 2026 → February 1, 2026	Phase 3	Train predictive model on full dataset, integrate trend-driven content ideatior	35	FUTURE
SPR-9	February 2, 2026 → February 15, 2026	Phase 3	Implement SHAP/LIME explainability layer, link predictions and content sugge	30	FUTURE
SPR-10	February 16, 2026 → March 1, 2026	Phase 4	Integrate UI, backend, M1, M2, M3 - initial full system testing	30	FUTURE
SPR-11	March 2, 2026 → March 15, 2026	Phase 4	Optimise engagement predictions, trend-to-idea mapping, recommendation	30	FUTURE
SPR-12	March 16, 2026 → March 29, 2026	Phase 4	Finalise Project Package Submission	30	FUTURE



RISKS AFFECTING THE PROJECT

Risk	Description	Impact	Likelihood	Mitigation/Control Measures	Contingency Plan
Learning curve for new technologies	Potential delays due to limited experience with FastAPI, Streamlit, SHAP, and TensorFlow/PyTorch or NLP libraries	High	High	Dedicate early sprints to setup and experimentation	Simplify architecture by using fewer frameworks
Data related and ethical risks	Poor data quality, API limits or bias could reduce accuracy and relevance or raise ethical issues	High	Medium	Use verified datasets, anonymise sensitive data, follow legal/ethical data policies	Replace live data with cached/synthetic alternatives
Feature creep/scope expansion	Risk of adding new advanced analytics before MVP completion	High	Medium	Enforce MVP scope and log future ideas in Notion for later development	Postpone all non-essential features until MVP completion
Model performance and explainability limitations	SHAP/LIME or predictive models may underperform	High	Medium	Validate against baseline metrics	Switch to simpler more interpretable models
Time management and scheduling risks	Illness, overlapping deadlines from other modules or sprint overruns	Medium	High	Add buffer time into sprints	Defer less critical tasks to later phases
Trend analysis reliability	NLP-based trend detection may produce irrelevant or low-quality due to noisy data	Medium	High	Apply filtering semantic similarity scoring and topic coherence measures	Instead of automated trend ideas, allow users to input custom trending topics
Recommendation logic accuracy	Content optimisation recommendations may not improve engagement if models are biased	High	Medium	Continues validation using historical data, implement A/B testing	Flag low confidence recommendations to users

LEGAL, SOCIAL, ETHICAL AND PROFESSIONAL CONSIDERATIONS

This project is defined as category 1, proof of concept prototype. It involved the development of an AI-driven prototype that doesn't handle sensitive or personal user data and presents low ethical and legal risk.

This project ensures compliance with all relevant data protection and intellectual property laws, including the UK Data Protection Act (2018) and the UK General Data Protection Regulation (UK GDPR) (Data Protection Act 2018). All datasets used will be either publicly available, licenced or anonymised to avoid handling personally identifiable information. Platform terms of services for TikTok, YouTube, Instagram and X will be strictly followed and data used for trend and performance analysis will be securely stored in encrypted local or cloud databases reducing risk of unauthorised aces or data leaks. Since the system is a proof-of-concept, it carries minimal commercial or legal exposure.

Ethical considerations are central to Creator Compass's design, particularly focused on fairness, transparency and explainability. The AI models will use techniques such as SHAP and Lime to clarify how predictions are made (Mosca *et al.*, 2022),ensuring interpretably for users. Balanced datasets will be used to minimise bias and results will be presented with clear disclaimers stating that outputs are only advisory.

Socially, the project seeks to empower creators by providing insight-driven tools that enhance creativity and performance. Creator Compass will contribute positively to the creator community. It encourages digital literacy by helping users to understand how data can be leveraged responsibly to improve engagement trends.

Professionally, the project adheres to the BCS code of conduct (BCS, 2022), ensuring honesty, integrity and respect for privacy. Development and documentation practices will follow professional standards for transparency, reproducibility and accountability. Ethical research protocols will also be maintained, with all datasets properly cited and model limitations clearly documented.

REFERENCES

Hase, P. and Bansal, M. (2020) 'Evaluating Explainable AI: Which Algorithmic Explanations Help Users Predict Model Behavior?' arXiv. Available at: https://doi.org/10.48550/arXiv.2005.01831.

Manic, M. (2024) 'Short-Form Video Content and Consumer Engagement in Digital Landscapes', *Bulletin of the Transilvania University of Brasov. Series V: Economic Sciences*, pp. 45–52. Available at: https://doi.org/10.31926/but.es.2024.17.66.1.4.

Mosca, E. *et al.* (2022) 'SHAP-Based Explanation Methods: A Review for NLP Interpretability', in N. Calzolari et al. (eds) *Proceedings of the 29th International Conference on Computational Linguistics. COLING 2022*, Gyeongju, Republic of Korea: International Committee on Computational Linguistics, pp. 4593–4603. Available at: https://aclanthology.org/2022.coling-1.406/ (Accessed: 5 November 2025).

Participation, E. (no date) *Data Protection Act 2018*. Statute Law Database. Available at: https://www.legislation.gov.uk/ukpga/2018/12/contents (Accessed: 5 November 2025).

Varga, S. and Cucu, E. (2025) *TikTok vs. Reels vs. Shorts (A Study by Socialinsider)*. Available at: https://www.socialinsider.io/blog/tiktok-vs-reels-vs-shorts/ (Accessed: 3 November 2025).

BCS (2022) BCS Code of Conduct. The Chartered Institute of IT. Available at: https://www.bcs.org/membership/become-a-member/bcs-code-of-conduct/ (Accessed 5 November 2025)

RESEARCH ETHICS CHECKLIST

A.1 If you answer YES to any of the questions in this block, your consultant/supervisor must have obtained approval for the project from an appropriate external ethics committee, and you need to have received written confirmation of this from him/her. Students cannot themselves apply for ethics approval in this case as the project is considered high risk". This type of research is not covered by City's process, and external approval from an appropriate institution is required.		*Delete as appropriate*
1.1	Does your research require approval from the National Research Ethics Service (NRES)?	**NO**
1.2	Will you recruit participants who are covered by the Mental Capacity Act 2005?	**NO**
1.3	Will you recruit any participants who are covered by the Criminal Justice System, for example, people on remand, prisoners and those on probation?	**NO**
A.2 If you answer YES to any of the questions in this block your consultant/supervisor must have obtained appropriate ethics committee approval		*Delete as appropriate*
2.1	Does your research involve participants who are unable to give informed consent? *For example, people who may have a degree of learning disability or mental health problem, that means they are unable to make an informed decision on their own behalf.*	**NO**
2.2	Is there a risk that your research might lead to disclosures from participants concerning their involvement in illegal activities?	**NO**
2.3	Is there a risk that obscene and or illegal material may need to be accessed for your research study (including online content and other material)?	**NO**
2.4	Does your project involve participants disclosing information about protected characteristics (as identified by the Equality Act 2010)? *For example: racial or ethnic origin; political opinions; religious beliefs; trade union membership; physical or mental health; sexual life; criminal offences and proceedings*	**NO**
2.5	Does your research involve you travelling to another country outside of the UK, where the Foreign & Commonwealth Office has issued a travel warning that affects the area in which you will study?	**NO**

	Please check the latest guidance from the FCO - http://www.fco.gov.uk/en/	
2.6	Does your research involve invasive or intrusive procedures? *These may include, but are not limited to, electrical stimulation, heat, cold or bruising.*	**NO**
2.7	Does your research involve animals?	**NO**
2.8	Does your research involve the administration of drugs, placebos or other substances to study participants?	**NO**
A.3 If you answer YES to any of the questions in this block, then unless you are applying to an external ethics committee or the Senate Research Ethics Committee (SREC), you must apply for approval from the Computer Science Research Ethics Committee (CSREC) through Research Ethics Online - https://researchmanager.city.ac.uk/. Depending on the level of risk associated with your application, it may be referred to the Senate Research Ethics Committee (SREC).		*Delete as appropriate*
3.1	Does your research involve participants who are under the age of 18?	**NO**
3.2	Does your research involve adults who are vulnerable because of their social, psychological or medical circumstances (vulnerable adults)? *This includes adults with cognitive and / or learning disabilities, adults with physical disabilities and older people.*	**NO**
3.3	Are participants recruited because they are staff or students of City, University of London? *For example, students studying on a particular course or module.* *If yes, then approval is also required from the Head of Department or Programme Director.*	**NO**
3.4	Does your research involve intentional deception of participants?	**NO**
3.5	Does your research involve participants taking part without their informed consent?	**NO**
3.5	Is the risk posed to participants greater than that in normal working life?	**NO**
3.7	Is the risk posed to you, the researcher(s), greater than that in normal working life?	**NO**
A.4 If you answer YES to the following question and your answers to all other questions in sections A1, A2 and A3 are NO, then your project is deemed to be of MINIMAL RISK. **If this is the case, then you can apply for approval through your supervisor under PROPORTIONATE REVIEW. You do so by completing PART B of this form.** **If you have answered NO to all questions on this form, then your project does not require ethical approval. You should submit and retain this form as evidence of this.**		*Delete as appropriate*
4	Does your project involve human participants or their identifiable personal data? *For example, as interviewees, respondents to a survey or participants in testing.*	**NO**